

A Multiform Optimization Framework for Multiobjective Feature Selection in Classification

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Abstract—Feature selection in machine learning as a key data processing technique has two conflicting goals: 1) minimizing the classification error rate and 2) minimizing the number of features selected. However, most of the existing multiobjective feature selection methods face the problems of easily falling into local optima and slow convergence by virtue of their problem characteristics, such as partially conflicting objectives and highly discontinuous Pareto fronts. To solve these problems, this article proposes a multiform optimization framework to solve a multiobjective feature selection task together with several auxiliary single-objective feature selection tasks in a multitask environment. The proposed framework uses the problem-solving experience of single-objective tasks to assist the multiobjective feature selection task in exploring more promising regions and accelerating the convergence speed. Specifically, a knowledge transfer strategy based on the search experience of different tasks is developed to accomplish multiform optimization. In addition, a diversity enhancement mechanism is presented to improve search ability in promising decision space areas by considering historical information about the population. In most cases, the experiment results on 27 datasets demonstrate that the proposed technique can uncover more diversified feature subsets on the Pareto front in less time than existing state-of-the-art methods.

Index Terms—Feature selection, multiform optimization, knowledge transfer, classification.

I. INTRODUCTION

CLASSIFICATION is a mainstream task in pattern recognition, which seeks to classify each instance into various groups according to the data presented by its features. It is challenging to perform fast and accurate classification of unknown data in datasets with hundreds or thousands of features [1]. This is because these datasets usually have numerous redundant and irrelevant features, which will reduce the accuracy of the classification [2] and increase the training time [3]. Therefore, a preprocessing technique for the dimensionality reduction of the data is required. Feature selection is one of the most popular research directions in classification [4], which can find feature subsets with stronger discrimination ability by removing redundant and irrelevant features from the complex and huge data [5], [6].

Generally, three procedures can be distinguished for the selection of features on the basis of different evaluation criteria. Filter-based, wrapper-based, and embedded-based are these three types of approaches [7]. The filter-based approach selects features based on the strength of the relationship between every feature and the objective variable, while a learner is used to evaluate the performance of feature subsets in the wrapper-based approach. Algorithm training and feature selection are performed simultaneously in the embedded-based method. The algorithm selects features based on their contribution to the model. In general, compared to filter-based and embedded-based methods, the wrapper-based method typically yields higher classification accuracy [8]. However, compared to filter-based and embedded-based approaches, it frequently works more slowly [9].

Evolutionary computation is well known for its powerful global search capability and is frequently employed to resolve multiobjective feature selection problems [10], [11], [12], [13], [14], [15]. For instance, Xue et al. [11] proposed a multiobjective binary feature selection method that integrates adaptive operator selection mechanisms. During the evolutionary process, this method uses a roulette wheel selection strategy to determine an appropriate crossover operator for generating the individual position of the next generation. However, it struggles with a hefty computational expense. The reason is that the probability of each crossover operator being

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selected needs to be calculated using the dominance relation matrix. Han et al. [16] proposed a reverse mutation operation based on feature frequencies. In addition, an adaptive environmental selection mechanism based on penalty-based boundary intersection parameters was designed to adjust the penalty value in the evolutionary process. However, this algorithm still faces the issue of falling into a local optimum since the feature interactions are not considered in the feature frequency ranking. Cheng et al. [13] proposed a variable-granularity method that aims to narrow the search space by using one granularity to contain multiple features. In evolution, refining the granularity allows the selection of key features. However, this method may face the problem of not finding a better Pareto front. This is because some useful features may be eliminated before refining the granularity. Therefore, the potential of evolutionary computation to solve the multiobjective feature selection problems remains to be further investigated.

Another important research area is feature selection as a single-objective optimization problem [17], [18], [19]. Researchers typically use a specific direction vector in such methods to regulate the ratio of the classification error rate to chosen features' number. A specific direction vector can be viewed as a subregion in the objective space, allowing the single-objective feature selection algorithm to explore deeply in the specified direction or subregion in order to quickly find promising feature subsets.

Multiform optimization [20], [21], [22], [23] as an emerging optimization paradigm is employed to improve the performance of global search by leveraging search experience from alternate formulations. The success of multiform optimization lies in capturing the different properties of the problem present in other alternate formulations to gain knowledge that can help to solve the original problem. Different from multitask optimization, multiform optimization focuses on a single target task with multiple forms. In this case, the search space is defined by the variables and the possible forms of the objective function. For instance, according to the problem characteristics, Yuan et al. [21] designed three reformulated helper multiobjective tasks to assist in solving the original complex multiobjective problem. Zhang et al. [22] constructed an auxiliary task by Gaussian process regression and then solved the original multiobjective optimization task together with the auxiliary task through a multiform optimization method. By exploiting knowledge sharing between original and auxiliary tasks, the auxiliary task is expected to provide helpful information for the evolution of the original task.

Motivated by the above, a new multiform evolutionary framework combining single and multiobjective formulations is proposed in this article to address complex multiobjective feature selection problems. In this proposed method, in addition to the original given multiobjective formulation, several auxiliary single-objective formulations with different weight vectors are also constructed to guide the multiobjective population to explore more promising regions. The auxiliary tasks and the original task are tackled concurrently, which can ensure the quality of knowledge transfer. To ensure similar evolutionary environments for both single-objective and multiobjective tasks, a decomposition-based framework

(MOEA/D) is employed as a vehicle for multiobjective tasks. It is noted that although MOEA/D decomposes the multiobjective problem into multiple subproblems to solve, it is different from the single-objective formulation in terms of environmental selection and search focus. These different forms of formulations have the same feature search space. The valid knowledge present in single-objective and multiobjective formulations is transferred to each other in the evolutionary process. It can not only enhance the population's convergence performance but also improve each individual's ability to jump out of a potential local optimum.

The main contributions of this article are summarized as follows.

- 1) A novel multiform multiobjective evolutionary feature selection framework (MFMOEA) is proposed to evolve a diverse set of nondominated feature combinations, which achieves superior performance in terms of approximation, accuracy, diversity, and computational cost. During the evolutionary process, auxiliary single-objective feature selection tasks can quickly identify promising regions in the decision space and guide the original multiobjective feature selection task to further explore promising regions, thereby accelerating the convergence of the multiobjective task.
- 2) An effective knowledge transfer strategy is developed for sharing effective feature combinations between the original multiobjective feature selection task and auxiliary single-objective tasks. Moreover, guided reproduction and similarity weight metrics are used to increase the probability of positive knowledge transfer. Due to the exchange of search experience between these two formulations, the population can more effectively maintain a balance between convergence and diversity in the evolutionary process.
- 3) A new diversity enhancement mechanism is designed to enhance the diversity of the population in the decision space. First, historical information of the population is utilized to detect duplicate offspring. Then, new offspring are generated to replace the duplicates by searching in promising regions, which enhances the exploration capability of the proposed method.

At the completion of our work, we find that a similar multiform approach (MFFS) was proposed by Jiao et al. [23]. In MFFS, two simplified single-objective feature selection tasks are used as auxiliary tasks for the original multiobjective feature selection task. However, the proposed framework in our study is actually different from MFFS. First, the motivations of single-objective tasks used in these two methods are different. MFFS is to supplement the diversity of solutions in the multiobjective task by the search of single-objective auxiliary tasks. But our algorithm utilizes single-objective tasks to accelerate the convergence of the multiobjective task. Second, the ways of generating auxiliary tasks are different. In our study, the number of generating auxiliary tasks is adjusted by an angle value, while MFFS has only two auxiliary tasks. Third, the way of knowledge transfer between auxiliary single-objective tasks and the original multiobjective task is different. The proposed knowledge transfer strategy in our study not

only considers the knowledge sharing between auxiliary tasks and the original task through parent selection and genetic operators, but also considers the transfer of elite solutions to more suitable tasks. However, MFFS considers the transfer of solutions only in environmental selection. According to the results of the experiments in Section V, our proposed algorithm performs better than MFFS.

II. RELATED WORK

In this section, we first introduce two forms of feature selection algorithms based on existing research and analyze their advantages and disadvantages. Then, the research motivation is clarified in detail.

A. Single-Objective Feature Selection

In single-objective feature selection problems, a fitness function is used to evaluate the performance of the selected feature subsets, which is usually a combination of the classification error rate and the number of selected features by weight. Researchers have proposed a lot of single-objective feature selection approaches in the past few years [17], [24], [25], [26], [27]. These approaches design a variety of search strategies to improve the chosen feature subsets' capacity for discrimination.

Designing effective search strategies to narrow the search space has become a popular method. For example, Unler and Murat [28] proposed a discrete single-objective particle swarm optimization algorithm with feature relevance weights computed to address feature selection. Adaptive selection and extended social learning strategies are designed to make the algorithm converge quickly to promising regions, so as to find excellent feature subsets.

It is another effective way to use multipopulation co-evolution or knowledge transfer between multitasks to solve the high-dimensional feature selection problem. Song et al. [29] combined co-evolution and single-objective algorithms. First, relevant features are divided into subspaces based on feature importance. Then, an initialization strategy based on feature sorting and an adaptive mechanism of particle change strategy are proposed to speed up the convergence. Evolutionary curves in results demonstrate that one advantage of the single-objective algorithm is fast convergence. Chen et al. [30] provided a multitask framework for single-objective feature selection. The original high-dimensional feature space is partitioned into many low-dimensional subspaces via this approach. Chen et al. proposed a multitask single-objective feature selection framework. In this algorithm, the original high-dimensional feature space is divided into multiple low-dimensional subspaces. An optimal feature subset is found as valid information searched in these low-dimensional subtasks is shared to facilitate evolution.

In summary, single-objective algorithms have the ability to discover helpful information in promising regions quickly. However, only one single solution can be outputted, which cannot provide decision makers with more alternative solutions.

B. Multiobjective Feature Selection

For a feature subset, reducing the number of selected features while reducing classification error rate is more considered as a multiobjective problem with two conflicting objectives [9], [31], [32]. These two objectives need to be minimized at the same time in the evolutionary process [33], as shown in

$$\begin{aligned} \min \quad & F(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x})) \\ \text{s.t.} \quad & \mathbf{x} \in \Omega \end{aligned} \quad (1)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_D)^T$ is a D -dimensional decision vector, and Ω is the decision space. $F(\mathbf{x})$ denotes the 2-D objective vector in feature selection. During the evolution, these two objectives are compared by dominance relationships, so a Pareto front can be obtained in the end [34]. Multiple solutions on the Pareto front can provide multiple feature subsets for users with different preferences. In the current research, the methods to solve the multiobjective feature selection problem can be roughly divided into the following four types.

Setting reference points or weight vectors is a common strategy to decompose a multiobjective problem into multiple subproblems. Nguyen et al. [35] proposed two reference point mechanisms based on the MOEA/D [36] framework. The search direction is decomposed using static reference points. Dynamic reference points are used exclusively for searching in areas with conflicting objectives. They help the algorithm get over difficulties brought on by extremely discontinuous Pareto fronts, imbalanced preferences and partially conflicting objectives in feature selection. Although the algorithm based on reference point decomposition can obtain a set of solutions of various subset sizes, it does not explore the decision space enough resulting in some feature proportions missing on the Pareto front. Thus, it falls into a local optimum.

Niche strategies can also be used to divide the evolving population into multiple subpopulations to address the lack of diversity of feature selection solutions, as well as to work toward finding more multimodal solutions. Wang et al. [32] proposed a multiobjective feature selection algorithm based on clustering to divide niches. Diversity is ensured by simultaneously considering the crowding distances in both decision space and objective space of solutions within subpopulations. In addition, an improved hypervolume (HV) contribution indicator and the nondominated solutions from all subarchives are used to lead the population to search more effectively and ensure convergence. Experiments show that this method can find better feature subset Pareto fronts and find more multimodal solutions. In [37], on the basis of the niche differential evolution algorithm, three enhancement strategies are performed to find multiple feature subsets with similar or same classification performance. First, the interaction variation combining niche and global information is designed to prevent offspring from falling into a local optimum. Second, the environmental selection is enhanced to identify more quasi-equal feature subsets while taking into account the proximity of the classification accuracy between the two feature subsets. Finally, a repair mechanism of quasi-equal subsets is designed to remove redundant features. However, the

niche-based multiobjective feature selection algorithm usually involves many sensitive parameters such as radius size and the number of niches. Their values significantly affect the effectiveness of the algorithm. Furthermore, determining the optimal value for each of them is difficult.

Shrinking the decision space is also an effective way to solve multiobjective feature selection problems. Feature importance ranking is a typical fundamental approach. In [38], the conditional entropy of each feature is calculated to reduce the probability of irrelevant features being selected. The larger the conditional entropy, the higher the correlation between the feature and the category. To produce a set of excellent initial solutions and accelerate the convergence, feature entropy is compared with a random number on each bit of each solution. Cheng et al. [39] designed a population steering matrix to reflect both the importance of features and the importance of individuals in the population. This matrix is used to reduce many unimportant and irrelevant features during evolution. But the ranking of features in one index does not reflect the interaction performance between them when the features are high dimensional. Moreover, it is necessary to find a combination of features with strong interaction to enhance classification accuracy.

Constructing or finding relevant feature selection tasks and then discovering valuable knowledge from them that facilitates the solution of the source task is an emerging approach [40]. Lin et al. [41] proposed a multitask feature selection algorithm framework that uses probabilistic models to extract knowledge from the source task. Common knowledge of related tasks is then found by analyzing the relationship between probabilistic models. Finally, the experimental findings demonstrate that applying this widespread knowledge can enhance the target task's feature selection effect. However, building highly relevant auxiliary tasks is a challenge in the first place. In addition, as data dimensions become tens of thousands, the usefulness of knowledge calculated by models will be reduced. This is easy to lead to the occurrence of negative transfer.

In summary, several nondominated feature subsets that can better satisfy the requirements of real-world applications are discovered by the multiobjective feature selection algorithm. However, the long time consumed and ease of getting into local optimal remain in such methods, especially when the feature space is enormous.

C. Motivation

The analysis of the above two sections shows that multiobjective feature selection still faces the problems of the long time consumed and easy to get into local optimal. However, the single-objective feature selection has the advantage of fast convergence, which can quickly find a promising region. Inspired by the idea of complementary advantages, we try to use the single-objective algorithm as auxiliary tasks to assist the multiobjective algorithm to converge quickly and find more diverse feature subsets. First, more promising regions in the unexplored decision space are found due to the fast convergence capability of the single-objective algorithm. Then, through knowledge transfer between the two

kinds of tasks, promising areas are provided to the original multiobjective task to enhance its exploration ability and diversity. Besides, it can speed up the convergence of the original multiobjective task and shorten the computation time. In Section III, the proposed approach is described in great depth.

III. PROPOSED METHOD

This section begins by describing how the auxiliary single-objective feature selection task is generated. Next, the proposed MFMOEA method's framework is presented. Finally, two key strategies of MFMOEA are described in detail.

A. Generation of Auxiliary Tasks

In the multiobjective feature selection task, minimizing the selected feature ratio and the classification error rate are two conflicting objectives. To better assist the original multiobjective feature selection task $\mathcal{T}_M$, this study intends to incorporate the above two objectives into the design of the auxiliary single-objective feature selection task $\mathcal{T}_S$. The main concept of the auxiliary task $\mathcal{T}_S$ construction is to generate single-objective auxiliary subtasks with different search directions. Their different preferences for two objectives can divide the objective space into several subregions. This provides useful knowledge for the search of the original task $\mathcal{T}_M$ in nearby subregions. This generation method not only shortens the time to find promising areas but also enhances the success rate of knowledge transfer between $\mathcal{T}_M$ and $\mathcal{T}_S$. The detailed steps for generating multiple auxiliary subtasks with different preferences for objectives are shown as follows. First, the selected feature ratio and the classification error rate are projected into a 2-D objective space, and the angle between these two objectives is 90° . Then, an angle value Angle is just set to control the number of generated auxiliary subtasks, calculated as $NT = \lfloor 90^\circ / \text{Angle} \rfloor - 1$. $\lfloor \cdot \rfloor$ is the symbol for rounding down. Next, the preferences of each auxiliary subtask for the selected feature ratio and classification error rate are calculated, respectively, denoted as $\alpha = \sin(h * \text{Angle})$ and $\beta = 1 - \sin(h * \text{Angle})$, where $h = \{1, 2, \dots, NT\}$. Finally, forming the fitness function corresponding to each auxiliary subtask, the formula is $fit_h = \alpha * f_1 + \beta * f_2$, where f_1 and f_2 represent the selected feature ratio and the classification error rate, respectively. The auxiliary subtasks with different weights based on the above steps have different search directions in the objective space. The search experience of these auxiliary subtasks can provide assistance to individuals with similar search directions in the original task $\mathcal{T}_M$.

Fig. 1 shows an example of generating auxiliary subtasks, which can be used to better understand the generation process of $\mathcal{T}_S$. Suppose $\text{Angle} = 30^\circ$, the number of generated auxiliary subtasks can be determined to be $NT = \lfloor 90^\circ / 30^\circ \rfloor - 1 = 2$. The preference for feature selection ratio and classification error rate is $(\alpha, \beta) = \{(0.5, 0.5), (0.866, 0.134)\}$. The single-objective feature selection subtasks are dispersed evenly across the objective space in this fashion. Different weight vectors show various preferences for two objectives during the convergence process. Therefore, auxiliary single-objective

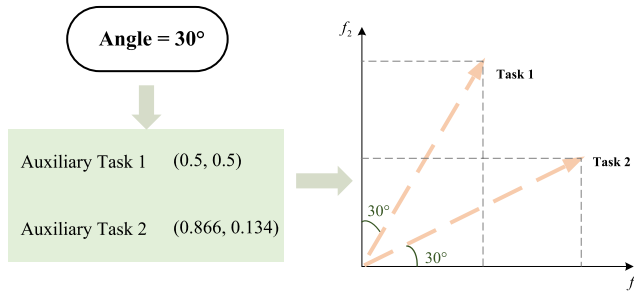


Fig. 1. Example is used to generate auxiliary tasks.

subtasks finally can find the corresponding different feature subsets. These diverse feature subsets containing some promising knowledge are migrated to the original multiobjective task $\mathcal{T}_M$. This can complement some missing solutions on the Pareto front and help original multiobjective task $\mathcal{T}_M$ jump out of the local optimum.

B. General Algorithm Framework

The proposed MFMOEA method's framework is described in Algorithm 1. First, several auxiliary single-objective subtasks are obtained according to the generation method in Section III-A (line 1). The original multiobjective task $\mathcal{T}_M$ is considered to be set up using an evolutionary algorithm framework based on decomposition. In this way, $\mathcal{T}_M$ and $\mathcal{T}_S$ have similar evolutionary environments. Second, half of the evaluated population is randomly assigned to $\mathcal{T}_M$, and the rest is assigned to $\mathcal{T}_S$ (lines 2–4). The reason is that two forms of tasks are required to take advantage of each other's strengths.

Third, a novel knowledge transfer approach combining implicit and explicit knowledge transfer is designed to share information between $\mathcal{T}_M$ and $\mathcal{T}_S$. Implicit knowledge transfer is used to guide the original multiobjective task $\mathcal{T}_M$ to produce promising offspring through achieving search experience from the auxiliary single-objective task $\mathcal{T}_S$ (line 6). Explicit knowledge transfer is used to retain the migrated elite solutions (line 10). Fourth, the offspring solutions of both form tasks are compared with the parent solutions to remove duplicate solutions in the decision space. The elite solutions are somewhat altered in accordance with the proposed diversity enhancement mechanism to produce some new solutions that will supplement the offspring population (line 7). If the termination criterion is satisfied, the proposed method outputs a set of nondominated feature subsets (lines 12 and 13); otherwise, the evolution will continue.

C. Knowledge Transfer Strategy

Knowledge transfer is the core part of the proposed multiform framework to realize knowledge sharing between the original multiobjective feature selection task $\mathcal{T}_M$ and the auxiliary single-objective feature selection task $\mathcal{T}_S$. To better share information between these two formulations, a new knowledge transfer strategy combining explicit and implicit transfer is designed in this study. The details are shown as follows.

Implicit Knowledge Transfer: It is critical to determine how to transfer implicit knowledge between $\mathcal{T}_M$ and $\mathcal{T}_S$ to help

Algorithm 1: General Framework of MFMOEA

Input: Population size NP , the original multi-objective task $\mathcal{T}_M$, the auxiliary single-objective task $\mathcal{T}_S$.

Output: Nondominated feature subsets.

- 1 Generate the weight vector for each individual in $\mathcal{T}_M$ and $\mathcal{T}_S$;
- 2 Randomly initialize population with NP individuals;
- 3 Evaluate the objective values of NP individuals and calculate the fitness values of the individuals in $\mathcal{T}_S$;
- 4 Assign $NP/2$ individuals to $\mathcal{T}_M$ and $\mathcal{T}_S$ respectively;
- 5 **while** the termination criterion is not met **do**
- 6 Select the parent individuals according to **Algorithm 2** and perform crossover and mutation operation;
- 7 Eliminate duplicate offspring and regenerate the promising offspring according to **Algorithm 4**;
- 8 Update the order of individuals in $\mathcal{T}_M$ and $\mathcal{T}_S$ according to their differences between the two objective values from the smallest to the largest;
- 9 Perform environmental selection by the Tchebycheff approach in $\mathcal{T}_M$ and by the order of fitness values in $\mathcal{T}_S$;
- 10 Perform explicit knowledge transfer according to **Algorithm 3**;
- 11 **end**
- 12 Perform nondominated sorting on the population;
- 13 Return nondominated feature subsets.

each other search for good feature subsets more effectively. In the framework of the genetic algorithm, parent selection and genetic operators determine information transfer between parents and offspring. Therefore, we use parent selection and genetic operator to implement implicit knowledge transfer between $\mathcal{T}_M$ and $\mathcal{T}_S$. The pseudocode of implicit knowledge transfer is shown in Algorithm 2. First, the neighborhood of each individual in $\mathcal{T}_M$ and $\mathcal{T}_S$ is arranged according to the similarity of weight vectors and objective values in the task where it is located, respectively. The reason for this approach in the former is that similar evolutionary environments are more likely to contain useful information. The latter does so because the size of similar feature subsets will favor convergence. The Euclidean distance matrix $\mathcal{D}_M$ between the weight vectors of each individual and other individuals in $\mathcal{T}_M$ is calculated (line 1). The closest T individuals are selected as the neighborhood of the current individual (line 3). Similarly, the neighborhood range of each individual in $\mathcal{T}_S$ can be calculated according to the objective values (lines 2 and 3). The neighborhood $\mathcal{N}_1$ of an individual is treated as a candidate pool for parent selection. Other individuals outside the neighborhood are called the external neighborhood $\mathcal{N}_2$ of each individual. Second, the elite solutions in the auxiliary single-objective task $\mathcal{T}_S$ are selected as part of the external neighborhood of individuals in $\mathcal{T}_M$ (line 5). At the same time, the nondominated solutions in the original multiobjective task $\mathcal{T}_M$ are added to the external neighborhood of each individual in auxiliary single-objective tasks $\mathcal{T}_S$ (line 6). The elite solutions

Algorithm 2: Implicit Knowledge Transfer

Input: Population $\mathcal{P}$, population size NP , weight vectors, objective values, multi-objective nondominated individuals, single-objective elite individuals, the original multi-objective task $\mathcal{T}_M$, the auxiliary single-objective task $\mathcal{T}_S$.

Output: The chosen parents $\mathbf{p}$.

```

1  $\mathcal{D}_M \leftarrow$  Calculate Euclidean distances among the weight
  vectors of individuals in  $\mathcal{T}_M$ ;
2  $\mathcal{D}_S \leftarrow$  Calculate Euclidean distances among the objective
  values of individuals in  $\mathcal{T}_S$ ;
3 Neighborhood  $\mathcal{N}_1$  of each individual  $\leftarrow$  Choose  $T$ 
  closest individuals base on  $\mathcal{D}_M$  or  $\mathcal{D}_S$ ;
4 Outer neighborhood  $\mathcal{N}_2$  of each individual  $\leftarrow$  Other
  individuals in the same task;
5 Add single-objective elite individuals to  $\mathcal{N}_2$  of each
  individual in  $\mathcal{T}_M$ ;
6 Add multi-objective nondominated individuals to  $\mathcal{N}_2$  of
  each individual in  $\mathcal{T}_S$ ;
7 for  $i = 1 : NP$  do
8   if  $Rand \leq CR$  then
9     randomly select an individual in  $\mathcal{N}_1$  as  $\mathbf{p}$ ;
10  else
11    randomly select an individual in  $\mathcal{N}_2$  as  $\mathbf{p}$ ;
12  end
13 end
14 Return  $\mathbf{p}$  of each individual.

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are the advanced knowledge carrier in the current generation. Third, the other parent $\mathbf{p}$ is selected from the neighborhood of each individual when the random value is less than CR (lines 8 and 9). Otherwise, an individual within the external neighborhood is selected as one of the parents (line 11). Finally, an offspring solution is generated according to the two parents by crossover operation and mutation operation. On the one hand, this implicit knowledge transfer can transfer the helpful feature combinations from auxiliary single-objective tasks $\mathcal{T}_S$ to the offspring of the original multiobjective task $\mathcal{T}_M$ to help it converge faster. On the other hand, advanced knowledge from $\mathcal{T}_M$ is also transferred to $\mathcal{T}_S$ so that they can search for some useful new feature subsets in the promising region.

Explicit Knowledge Transfer: The auxiliary single-objective feature selection task $\mathcal{T}_S$ has the benefit of quick convergence, but it has a single search direction. Explicit knowledge transfer can transfer the nondominated solutions $\mathcal{F}$ in $\mathcal{T}_M$ to the auxiliary single-objective feature selection task $\mathcal{T}_S$, which can effectively improve the subpopulations' diversity in $\mathcal{T}_S$. Algorithm 3 presents the pseudocode of explicit knowledge transfer. First, the Euclidean distance between the weight vectors of each nondominated solution and the weight vector of each auxiliary single-objective task $\mathcal{T}_S$ is calculated (line 3).

This is because the Euclidean distance, which is easy to calculate, can more accurately depict how different two vectors are when all of the weight variables are between $[0, 1]$. Second, the auxiliary single-objective task $\mathcal{T}_{S_c}$ closest

Algorithm 3: Explicit Knowledge Transfer

Input: Population $\mathcal{P}$, objective value matrix $\mathcal{V}$ of $\mathcal{P}$, the multi-objective nondominated individual set $\mathcal{F}$, the multi-objective task weight set $\mathcal{W}_M$, all single-objective subtasks' weight set $\mathcal{W}_S$.

Output: Pop , Obj .

```

1  $M \leftarrow$  Calculate the number of individuals in  $\mathcal{F}$ ;
2 for  $i = 1 : M$  do
3   Calculate Euclidean distances between the weight
    vector of  $\mathcal{F}_i$  and each weight vector in  $\mathcal{W}_S$ ;
4    $\mathcal{T}_{S_c} \leftarrow$  The single-objective subtask with the smallest
    Euclidean distance;
5    $\mathbf{u} \leftarrow$  Find the individual at the bottom of the fitness
    ranking in  $\mathcal{T}_{S_c}$ ;
6    $\mathcal{P}(\mathbf{u}) \leftarrow \mathcal{P}(\mathcal{F}_i)$ ;
7    $\mathcal{V}(\mathbf{u}) \leftarrow \mathcal{V}(\mathcal{F}_i)$ ;
8 end
9 Return  $\mathcal{P}$  and  $\mathcal{V}$ .

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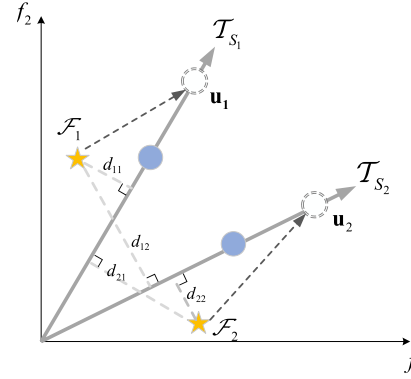


Fig. 2. Example of replacing the worst solution in explicit knowledge transfer.

to the current nondominated solution is found according to the weight distance (line 4). Finally, the worst solution $\mathbf{u}$ based on fitness ranking in $\mathcal{T}_{S_c}$ is replaced by the current nondominated solution (lines 5–7). Fig. 2 presents an example of explicit knowledge transfer. First, the Euclidean distances (d_{11} and d_{12}) between the nondominated individual $\mathcal{F}_1$ and all auxiliary single-objective tasks are computed, as well as the distances (d_{21} and d_{22}) between $\mathcal{F}_2$ and all auxiliary single-objective tasks. Second, the auxiliary tasks closest to nondominated individuals $\mathcal{F}_1$ and $\mathcal{F}_2$ are $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$, respectively, based on d_{11} being smaller than d_{12} and d_{22} being smaller than d_{21} . Then, $\mathcal{F}_1$ and $\mathcal{F}_2$ replace $\mathbf{u}_1$ and $\mathbf{u}_2$, respectively. The reason is that replacing the worst solution generally does not lose a potentially promising feature subset. Moreover, the nondominated solution can often become the new optimal or suboptimal solution in the auxiliary single-objective task.

D. Diversity Enhancement Mechanism

In the multiobjective feature selection, the existence of numerous duplicate solutions may lead to following problems. First, these duplicate solutions are evaluated many times, resulting in a long computation time and a waste of computing

Algorithm 4: Diversity Enhancement Mechanism

Input: Parent population $\mathcal{P}$, offspring population $\mathcal{O}$, local search solution set $\mathcal{L}$, population size NP .
Output: Offspring population $\mathcal{O}$.

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1  $\mathcal{O}_u \leftarrow$  Find unique offspring individuals in  $\mathcal{P} \cup \mathcal{O} \cup \mathcal{L}$ ;
2 if  $|\mathcal{O}_u| \geq NP$  then
3    $\mathcal{O} = \mathcal{O}_u(1 : NP)$ ;
4 else
5   Execute a two-stage search operation;
6   Stage1:
7      $\mathcal{E} \leftarrow$  Perform nondominated sorting in elite individuals;
8      $\mathcal{T}_{S_e} \leftarrow$  Find single-objective subtasks where single-objective efficient individuals are located;
9      $\mathcal{R}_S \leftarrow$  Select individuals located in the top half of  $\mathcal{T}_{S_e}$  according to the fitness rank;
10    Randomly invert dimensions for each individual in  $\mathcal{R}_S$ ;
11    Stage2:
12    Add or remove features to certain feature proportions of feature subsets in  $\mathcal{E}$  according to Eq. 2;
13     $\mathcal{L} \leftarrow$  Combine the individuals obtained by Stage1 and Stage2;
14     $\mathcal{O}_n = \mathcal{O}_u \cup \mathcal{L}$ ;
15     $\mathcal{O} = \mathcal{O}_n(1 : NP)$ ;
16 end
17 end
18 Return  $\mathcal{O}$  and  $\mathcal{L}$ ;
```

resources. Second, duplicate feature subsets make the algorithm susceptible to settling into a local optimum by reducing the diversity of possible solutions. For these reasons, we propose a diversity enhancement mechanism. It uses the historical information of the previous generation's search to eliminate duplicate solutions and regenerate the promising offspring to explore new search directions.

Algorithm 4 shows the detailed steps of our proposed mechanism. First, $\mathcal{P}$, $\mathcal{O}$, and $\mathcal{L}$ of the previous generation are compared to remove duplicate feature subsets. An offspring population $\mathcal{O}_u$ that does not duplicate the previous population is obtained (line 1). The number of individuals in $\mathcal{O}_u$ is also calculated (line 2). Second, the first NP solutions are taken as the offspring population when the number of unduplicated offspring meets the population size (lines 3). Otherwise, the offspring population should be complemented. A two-stage search operation is performed with auxiliary single-objective tasks to generate promising complementary solutions (lines 6–12). It is worth noting that the lack of exploration of promising feature combinations and the lack of some feature proportions on the Pareto front lead to the local optimization of the multiobjective feature selection algorithm. Therefore, the search operation is divided into two stages to solve these two problems separately.

For the former problem, the direction of exploration is first derived from the elite solutions of the two forms of task. A new

nondominated solution set $\mathcal{E}$ is obtained by performing a non-dominated sorting on the elite solutions in the multiobjective $\mathcal{T}_M$ and single-objective $\mathcal{T}_S$ tasks (line 7). If the solutions in $\mathcal{T}_S$ can be retained after sorting, it means that auxiliary single-objective tasks have searched for more effective feature combinations in the decision space. Then, the individuals of the subtask $\mathcal{T}_{S_e}$, where the effective solution is located, are also considered to be worth exploring (line 8). Thus, some dimensions of top-ranked individuals in $\mathcal{T}_{S_e}$ are randomly inverted (lines 9 and 10). The newly generated solutions retain promising features yet are less likely to fall into local optima. In this way, adding them as supplementary solutions to the offspring population $\mathcal{O}_u$ will play a more effective role than randomly generating solutions. The operations of stage 1 not only make full use of the search experience of all tasks to help accelerate the convergence of the original multiobjective task, but also improve the diversity of the promising feature subsets.

For the latter problem, it is necessary to add some new solutions to make up for the Pareto front with some feature ratios lost. To ensure that these new solutions are preserved on the Pareto front with high probability, nondominated feature subsets of all subtasks are considered to be adjusted to obtain the desired specific subset size. In addition, to identify which features can be deleted or added, the mutual information (MI) between features and categories is also assessed. Equation (2) displays the calculation formula for the MI that is employed

$$MI(x, y) = \sum p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \quad (2)$$

where x represents a feature and y represents a label, $p(x, y)$ stands for the joint probability of x and y , while $p(x)$ and $p(y)$ denote the marginal probabilities of x and y . The association between the feature and the category is greater the higher the value computed in this manner. Specifically, after the nondominated sorting of solutions in $\mathcal{T}_M$ and $\mathcal{T}_S$, it is found that the obtained new Pareto front is lost solutions with some feature ratios compared with the previous Pareto front of $\mathcal{T}_M$. Nevertheless, it is still possible to find potential nondominated solutions in the decision space to make up for the Pareto front with lost ratios. For each solution in $\mathcal{E}$, if the ratio of selected features is more than the lost ratio, the selected features which are at the end of MI rank will be deleted until the lost feature ratio is reached (line 12). On the contrary, if the selected feature ratio is less than lost ratio, the unselected features located in front of the ranking will be added until the lost feature ratio is reached (line 12).

As shown in Fig. 3, when the lost ratio is 0.4, individuals with only two features selected need to be generated. For the individual with three features selected, the fourth feature with the lowest MI value is removed. For the individual with one feature selected, the fifth feature with the second highest MI value is added. Finally, the solutions from the above two-stage operation are combined into one set $\mathcal{L}$ (line 13). The unduplicated offspring are then mixed with $\mathcal{L}$ to create a new population of offspring (line 14). To create the offspring population that needs to be evaluated in the current generation, the first NP solutions in the new offspring population are truncated (line 15).

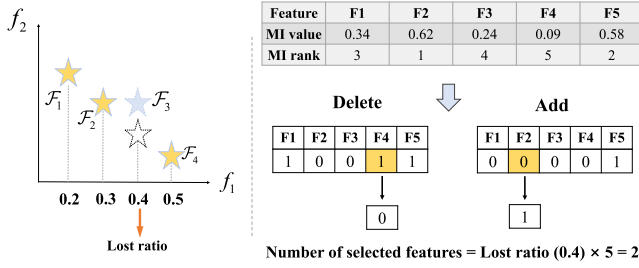


Fig. 3. Example of adjusting to the lost ratio in Stage2.

In summary, the duplicate feature subsets are replaced by promising solutions. The effectiveness of searches in decision space is greatly improved. The diversity of solutions on the Pareto front is also significantly enhanced.

E. Computational Complexity

The computational cost of MFMOEA comes mainly from the reproduction operation and the nondominated sorting operation. Specifically, first, the computational complexities of both the parent selection and genetic operators are $O(NP/2)$ in each task subpopulation, respectively. And, NP is the population size. Second, the worst-case computational complexity of the diversity enhancement mechanism is $O(E_M \cdot E_S)$, where E_M and E_S are the numbers of nondominated solutions in the multiobjective and single-objective tasks, respectively. Third, the computational complexity of the nondominated ranking of the two task populations separately is $O(2 \cdot 2 \cdot (NP/2)^2)$. Finally, the computational complexity of environmental selection is $O(NP/2 + N_W)$, where N_W is the number of different weight vectors in auxiliary single-objective tasks. Therefore, one generation's overall complexity of MFMOEA is $2 \cdot O(NP/2 \cdot 3) + O(E_M \cdot E_S) + O(2 \cdot M \cdot (NP/2)^2) + O(NP/2 + N_W) = O(NP^2)$, where M is the number of objectives.

However, MFMOEA is able to discover feature subsets with fewer features. Its computation time is shorter than NSGA-II and MOEA/D. This is verified in Section V-D.

IV. EXPERIMENTAL SETUPS

A series of experiments have been carried out to confirm the viability of the proposed feature selection method. The datasets under investigation, the compared algorithms and the parameter settings are all presented in this section.

A. Datasets

All 27 datasets used in the experiments are based on real-world data and downloaded from <https://archive.ics.uci.edu/ml/datasets.php>. As shown in Table I, the datasets are both low-dimensional and high-dimensional, with the specific number of features ranging from 8 to 11 340. Moreover, two-class problems and multiclass problems are included. They come from fields as diverse as disease and gene expression (Hepatitis, Prostate, Leukemia1, Leukemia2, and GLIOMA), physical chemistry (Sonar and Ionosphere) and speech recognition (Isolet5). It is worth noting that the ratio of the number of samples to the number of features for each dataset is calculated and presented in Table I. The value of this ratio

TABLE I
BASIC INFORMATION OF 27 DATASETS

No.	Datasets	#Features	#Classes	#Instances	#Instance/Feature
1	Abalone	8	3	4177	522.13
2	Wine	13	3	178	13.69
3	Australian	14	2	690	49.29
4	Zoo	16	7	101	6.31
5	Climate	18	2	540	30.00
6	Vehicle	18	4	752	41.78
7	Hepatitis	19	2	155	8.16
8	SPECTF Heart	22	2	267	12.14
9	German	24	2	1000	41.67
10	Flags	28	7	194	6.93
11	Ionosphere	34	2	351	10.32
12	Spambase	57	2	4601	80.72
13	Sonar	60	2	208	3.47
14	Optical	64	10	1797	28.08
15	Hill Valley	100	2	606	6.06
16	Arrhythmia	279	13	452	1.62
17	Madelon	500	2	2600	5.21
18	Isolet5	617	26	1559	2.53
19	SRBCT	2308	4	83	0.04
20	GLIOMA	4434	4	50	0.01
21	Leukemia1	5327	3	72	0.01
22	DLBCL	5469	2	77	0.01
23	Brain1	5920	5	90	0.02
24	Carcinom	9182	11	174	0.02
25	Prostate	10509	2	102	0.01
26	Leukemia2	11225	3	72	0.01
27	CLLSUB	11340	3	111	0.01

covers a particularly wide range, where it ranges from 0.01 to 522.13. These increase the difficulty of the algorithm in selecting key features.

Therefore, the datasets used in this experiment are sufficient to test the universality and effectiveness of each competing algorithm on classification. In our experiments, the classification performance of the proposed approach is evaluated using datasets 1–18. Datasets 19–27 are used to test the algorithm's scalability in high-dimensional feature selection problems.

B. Compared Algorithms and Experiment Settings

Six advanced compared algorithms are selected for comparison with MFMOEA. They are NSGA-II [42], MOEA/D [36], MOEAD-PFE [43], DAEA [9], SM-MOEA [39], and MFFS [23]. Among them, NSGA-II is the most classical algorithm that updates populations based on dominance relationships and usually measures the crowding distance to maintain population diversity. MOEA/D is a decomposition-based algorithm that assigns a different weight vector to each individual in the population, and uses the Tchebycheff approach to converge. MOEAD-PFE is a variant of MOEA/D. It uses a nondominated solution set and an extended set of weight vectors to estimate the distribution of a true Pareto front. The weight vectors are then adaptively adjusted. DAEA performs an iterative analysis of solutions in the decision and objective space, coupled with diversity-based environmental selection strategies. In SM-MOEA, a steering matrix that considers both features' importance and individuals' importance is used to guide to select effective and small feature subsets. MFFS uses a multiform framework similar to ours. But the two main strategies are completely different. These algorithms are selected as compared algorithms to assure the fairness of the

experiments since they have solid theoretical foundations and exhibit good performance in feature selection. Bold numbers indicate that the algorithm corresponding to them produced the best results.

Each dataset is split into a training set and a test set according to the sample number of various labels in the proportions of 80% and 20% to show the validity of the classification results [44]. The 5-fold cross-validation [45] is employed to assess the classification effect and avoid the effect of feature data bias during the training phase. The classification error rate is calculated using the straightforward and efficient KNN classifier. To balance efficiency and accuracy [9], the value of k in the KNN classifier is set to 5.

Two performance metrics, HV [46] and inverse generation distance (IGD) [47], are utilized in this study to assess the performance of the seven multiobjective algorithms. When calculating HV, it is guaranteed that both objective values are normalized to [0, 1]. The reference point is set to (1, 1) [9]. When calculating IGD, it is different from benchmark function problems. The search space of feature selection is composed of many dimensional features, so it is treated as a discrete problem. Moreover, the number of all possible feature combinations is 2^n leading to a search space that is too large, so that the true Pareto front is difficult to be found. Therefore, we first collected the nondominated solution sets for 30 runs of seven algorithms on each dataset and combined them into a large nondominated front by nondominated sorting. This large front is used as the reference front for computing IGD values and is assumed to be the roughly true front on this dataset. In general, the algorithm performs better when the HV value is higher; the method performs better when the IGD value is lower.

To determine if there is a significant performance difference between MFMOEA and the other six evaluated algorithms, the Wilcoxon test is also utilized, with a significance threshold of 0.05. According to the symbols “+,” “-,” and “=” the proposed MFMOEA algorithm is considerably superior to, inferior to, and comparable to the compared algorithms, respectively. To put it another way, the more + attached to other algorithms, the better the proposed algorithm works.

C. Parameter Settings

In this work, all experiments are implemented based on MATLAB. It is worth mentioning that these six compared algorithms are all derived from an open-source platform, PLAT-EMO [48]. This can guarantee the authenticity and validity of this experiment. To exclude the effect of randomness, each compared algorithm is independently run 30 times on each dataset. For different algorithms on the same dataset, the same initial random seed is used. The specific parameters of all compared algorithms can be found in their respective literature [9], [23], [36], [39], [42], [43]. To ensure the efficiency and diversity, the population size N of each algorithm is set to 200 and the maximal generation is set to 100.

In MFMOEA, the size of each individual's neighborhood is set to 10% of the population size for each task, as in the general setting in [34]. The probability CR of selecting a parent

TABLE II
SENSITIVITY ANALYSIS OF HV METRICS ON 12 REPRESENTATIVE DATASETS WITH DIFFERENT ANGLE VALUES

Dataset \ Angle (°)	5	10	15	30	45
Wine	0.8942	0.8890	0.8915	0.8903	0.8873
Zoo	0.8526	0.8534	0.8514	0.8491	0.8477
Hepatitis	0.8428	0.8415	0.8413	0.8407	0.8418
SPECTF Heart	0.7397	0.7362	0.7372	0.7335	0.7322
Ionosphere	0.9314	0.9316	0.9292	0.9291	0.9283
Sonar	0.8681	0.8700	0.8581	0.8484	0.8458
Optical	0.8823	0.8779	0.8838	0.8788	0.8790
Hill Valley	0.5963	0.5968	0.5890	0.5930	0.5969
Arrhythmia	0.6926	0.6936	0.6957	0.6904	0.6944
SRBCT	0.9881	0.9660	0.9715	0.9581	0.9604
DLBCL	0.9998	0.9922	0.9994	0.9935	0.9807
Leukemia2	0.9598	0.9430	0.9665	0.9783	0.9601
Friedman's rank	2.3333	3.2917	2.8333	4.5	4.5417

inside the neighborhood is set to 0.8, which is similar to the conventional setting in [9]. Furthermore, only one key parameter is the angle value Angle. This parameter is used to adjust the number of auxiliary single-objective subtasks with different weight vectors. Twelve datasets with a wide coverage of the ratio values of feature number to sample size are selected as representative datasets for the parameter sensitivity analysis experiments. As shown in Table II, the several different angle values tested are all factors of 90. These are sufficient to test the effect of angle value on algorithm performance. For each angle value, 30 experiments are performed on each dataset and the mean values of HV metrics were recorded. The results of Friedman's test shows that MFMOEA performs best when the angle value is 5. At this time, there are 17 different weight vectors in the entire single-objective subpopulation. And, the total single-objective subpopulation is divided equally into the individuals occupied by each weight vector. In addition, the crossover rate in each compared algorithm is set to 1 [9], which means that crossover must occur during reproduction. The specific features of the crossover are randomly determined. The probability of mutation is set to the inverse of the number of selected features. And, the features of mutation are also randomly selected. This means that the number of mutated features is related to the size of the feature subset, thereby ensuring the effectiveness of the mutation operation.

V. EMPIRICAL PERFORMANCE ANALYSIS

Several experiments are designed to demonstrate the advancement and effectiveness of the proposed MFMOEA in the feature selection problem compared to other algorithms. In addition, some detailed analyses are conducted to help understand the intrinsic mechanism of MFMOEA.

A. Nondominated Front Distribution Analysis

In order to intuitively reflect the advantages of MFMOEA's convergence and diversity, the median Pareto fronts of all compared algorithms on nine datasets with different dimensions are presented in Fig. 4. The chosen nine data sets are representative in low or high dimensions. The median value

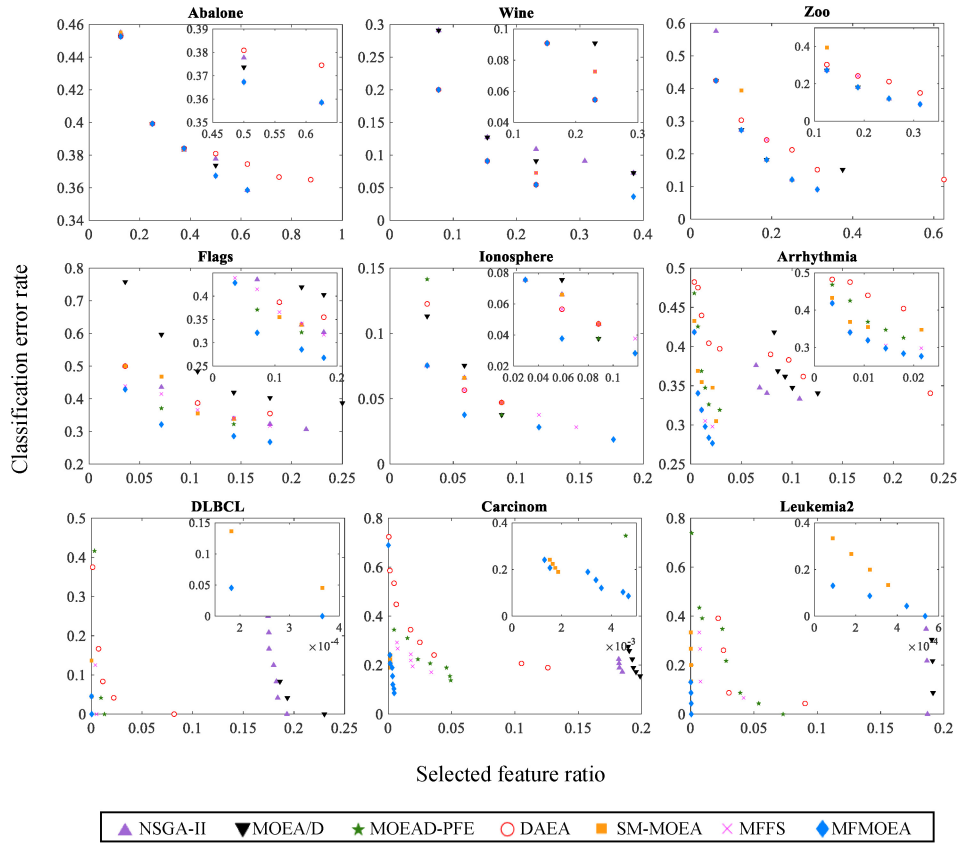


Fig. 4. Distributions of nondominated solutions obtained by each algorithm on test sets in terms of median HV value.

is found in the ranking of HV values from 30 independent experiments. The plotted Pareto front for this experiment is called the median Pareto front. It can reflect the general effect of the algorithm. First, Fig. 4 shows that the front found by MFMOEA has the best classification accuracy compared with other algorithms. Even on high-dimensional datasets, such as DLBCL and Leukemia2, where the ratios of the number of samples to the number of features are 0.01, feature subsets with the 0 classification error rate can be found. And, the number of features selected is also the least. It proves that our MFMOEA achieves a better exploration of the search space. Second, Pareto fronts found by MFMOEA are the most comprehensive and complete compared with Pareto fronts of other compared algorithms. The reason is that the diversity enhancement mechanism can use MI and elite solutions to repair the missing Pareto front and find more new non-dominated solutions. Finally, it is worth noting that MFPS, which is similar to our framework, is significantly less convergent than MFMOEA. Meanwhile, our algorithm can better solve high-dimensional problems compared with other algorithms. Specifically, MFMOEA can filter out a large number of redundant features and select only a few key features to achieve high classification accuracy.

B. MFMOEA Versus Other Compared Algorithms

Table III shows the mean value, standard deviation, rank sum test, and Friedman's test results of IGD value of 7 algorithms. It can be seen that Friedman's rank of MFMOEA is

the first. Specifically, MFMOEA can completely beat the three compared algorithms (NSGA-II, MOEA/D, and SM-MOEA) on all datasets in terms of IGD metric. In addition, the significance test results show that MFPS is significantly worse than MFMOEA on most datasets. These indicate that MFMOEA can find feature subsets with smaller sizes and higher classification accuracy on most data sets. The above phenomena show that the proposed knowledge transfer strategy makes full use of the fast convergence of auxiliary single-objective tasks to deliver effective feature combinations to the original multiobjective task. In our multiform framework, multiobjective, and single-objective tasks achieve knowledge sharing and co-evolution to find feature subsets with better classification performance. Moreover, the diversity enhancement mechanism makes MFMOEA fully explore the decision space and exploit elite feature combinations with a small number of features in promising areas.

It can be seen from Table IV that MFMOEA is significantly optimal based on the results of Friedman's test. This is mainly because the proposed diversity enhancement mechanism can detect and make up for some missing feature proportion points on the Pareto front. Then, nondominated feature subsets with better diversity are found in the objective space. It is worth noting that MFMOEA is superior to other compared algorithms in terms of HV on high-dimensional datasets with ratios of the samples' number to the features' number of 0.01, 0.02, and 0.04. It can be found that our proposed framework overcomes the challenge of small sample

TABLE III
MEAN IGD PERFORMANCE ON TEST DATA

Dataset	NSGA-II	MOEA/D	MOEA-PFE	DAEA	SM-MOEA	MFPS	MFMOEA
Abalone	2.40e-03±4.34e-03(+)	1.28e-02±2.40e-04(+)	1.26e-03±3.81e-03(+)	2.09e-03±1.76e-18(+)	2.65e-02±6.91e-03(+)	5.66e-02±2.12e-17(+)	1.11e-04±1.72e-04
Wine	7.05e-02±7.27e-03(+)	8.53e-02±1.63e-02(+)	5.84e-02±1.91e-02(+)	4.24e-02±1.69e-02(+)	5.14e-02±1.54e-02(+)	4.41e-02±7.31e-18(+)	3.37e-02±1.42e-02
Australian	5.98e-02±1.41e-17(+)	8.31e-02±3.06e-02(+)	5.48e-02±8.57e-03(+)	5.39e-02±1.42e-02(=)	6.95e-02±6.77e-03(+)	4.23e-02±0.00e+00(+)	3.52e-02±4.71e-03
Zoo	3.87e-02±2.05e-02(+)	4.67e-02±1.44e-02(+)	1.32e-02±4.02e-03(+)	5.01e-02±9.86e-03(+)	6.85e-02±1.08e-02(+)	2.93e-02±8.64e-03(+)	1.75e-03±4.54e-03
Climate	5.91e-03±3.31e-03(+)	2.21e-02±1.04e-02(+)	2.36e-03±1.61e-03(+)	7.18e-03±2.65e-03(+)	2.58e-03±1.70e-03(+)	2.95e-02±1.06e-17(+)	1.18e-04±2.58e-04
Vehicle	1.04e-02±2.84e-03(+)	2.87e-02±2.92e-03(+)	5.39e-03±2.78e-03(=)	6.26e-02±2.21e-02(+)	2.18e-02±7.34e-03(+)	3.04e-02±3.38e-03(+)	5.27e-03±2.79e-02
Hepatitis	1.59e-01±3.32e-02(+)	7.48e-02±1.81e-02(=)	7.81e-02±2.29e-02(=)	7.75e-02±1.38e-02(=)	9.32e-02±1.00e-02(+)	8.86e-02±1.58e-03(+)	7.43e-02±2.33e-02
SPECTF Heart	4.79e-02±2.38e-04(+)	2.01e-02±2.88e-02(+)	3.53e-03±1.15e-03(-)	5.14e-04±4.30e-04(-)	1.06e-01±1.95e-02(+)	4.04e-02±6.78e-04(+)	1.08e-02±1.72e-02
German	4.29e-04±9.99e-04(+)	6.33e-04±1.24e-03(+)	1.11e-04±2.85e-05(+)	5.62e-05±2.56e-05(+)	1.07e-04±5.43e-06(+)	6.79e-04±1.17e-07(+)	1.94e-05±5.74e-06
Flags	5.77e-02±3.56e-03(+)	9.80e-02±2.82e-17(+)	4.50e-02±7.29e-03(+)	8.71e-02±4.07e-03(+)	7.24e-02±3.59e-03(+)	1.80e-02±8.47e-17(+)	2.01e-03±2.96e-03
Ionosphere	2.77e-02±1.04e-02(+)	3.06e-02±1.62e-03(+)	3.14e-02±2.30e-03(+)	4.45e-02±8.24e-03(+)	3.13e-02±6.25e-03(+)	3.13e-02±1.59e-03(+)	3.10e-03±7.66e-03
Spambase	2.79e-02±1.07e-02(+)	8.92e-02±4.23e-17(+)	3.24e-02±5.35e-03(+)	5.84e-02±1.49e-02(+)	2.67e-02±2.15e-03(+)	2.35e-03±1.30e-03(=)	2.58e-03±1.40e-03
Sonar	7.83e-02±1.34e-02(+)	8.68e-02±1.22e-02(+)	4.40e-02±6.93e-03(+)	6.73e-02±1.55e-02(+)	6.51e-02±1.24e-02(+)	3.84e-02±2.75e-03(+)	3.71e-02±1.12e-02
Optical	7.14e-02±1.72e-02(+)	1.30e-01±8.47e-17(+)	4.69e-02±2.29e-03(+)	6.65e-02±3.18e-03(+)	6.81e-02±1.65e-02(+)	3.49e-03±1.26e-03(-)	2.98e-02±4.11e-03
Hill Valley	3.11e-02±1.13e-02(+)	5.58e-02±1.99e-02(+)	3.28e-02±5.32e-03(+)	3.71e-02±1.69e-02(+)	2.31e-02±2.30e-03(+)	2.19e-02±1.20e-03(+)	1.12e-02±7.05e-03
Arrhythmia	9.09e-02±1.14e-02(+)	1.10e-01±1.08e-02(+)	4.18e-02±1.18e-02(+)	9.37e-02±1.27e-02(+)	4.52e-02±1.47e-02(+)	9.44e-02±1.41e-02(+)	9.43e-03±6.88e-02
Madelon	1.48e-01±7.69e-03(+)	1.68e-01±7.92e-03(+)	7.13e-02±7.77e-03(+)	1.20e-01±9.24e-03(+)	4.40e-02±1.50e-02(+)	2.08e-02±1.54e-03(-)	1.78e-01±1.56e-02
Isolet5	4.51e-01±2.36e-02(+)	4.86e-01±2.46e-02(+)	5.03e-02±1.16e-03(+)	7.38e-02±4.88e-02(+)	3.53e-02±3.75e-02(+)	1.17e-02±8.26e-03(+)	1.89e-02±3.88e-03
SRBCT	1.86e-01±1.06e-02(+)	2.01e-01±8.27e-03(+)	3.77e-02±2.83e-03(+)	6.58e-02±1.29e-02(+)	7.75e-02±4.86e-02(+)	7.60e-02±3.25e-03(+)	1.58e-02±8.67e-03
GLIOMA	2.17e-01±1.10e-02(+)	2.21e-01±2.47e-03(+)	4.23e-02±1.46e-02(-)	4.83e-02±1.13e-02(-)	5.51e-02±2.89e-02(=)	5.14e-02±6.36e-03(+)	6.51e-02±2.55e-02
Leukemia1	2.21e-01±1.06e-02(+)	2.19e-01±1.27e-02(+)	9.61e-02±2.09e-02(+)	1.09e-01±4.84e-04(+)	9.51e-02±5.32e-02(+)	4.88e-02±6.43e-02(+)	9.05e-03±2.13e-02
DLBCL	1.93e-01±2.87e-03(+)	2.02e-01±5.37e-03(+)	2.35e-02±6.37e-03(+)	8.40e-02±1.58e-02(+)	3.00e-02±2.74e-02(+)	1.93e-02±1.02e-02(=)	1.91e-02±1.27e-02
Brain1	2.91e-01±1.04e-02(+)	2.81e-01±1.13e-16(+)	1.99e-01±1.76e-02(+)	2.02e-01±1.18e-03(+)	1.96e-01±4.73e-02(+)	1.68e-01±4.61e-02(+)	1.10e-01±2.47e-02
Carcinom	2.23e-01±3.85e-03(+)	2.17e-01±5.65e-17(+)	7.06e-02±7.20e-03(+)	7.77e-02±1.00e-02(+)	7.60e-02±2.63e-02(+)	7.88e-02±3.18e-02(+)	2.00e-02±1.02e-02
Prostate	2.07e-01±5.60e-03(+)	1.05e-01±2.82e-17(+)	4.39e-02±7.89e-03(+)	8.21e-02±1.25e-02(+)	5.92e-02±2.27e-02(+)	5.42e-02±3.15e-02(=)	3.99e-02±1.77e-02
Leukemia2	2.34e-01±2.55e-02(+)	1.96e-01±1.13e-16(+)	1.08e-01±3.19e-02(+)	1.32e-01±3.48e-02(+)	1.01e-01±6.81e-02(+)	4.96e-02±4.33e-02(=)	4.05e-02±3.15e-02
CLLSUB	2.62e-01±1.36e-02(+)	2.31e-01±1.13e-16(+)	5.33e-02±1.98e-02(=)	1.27e-01±1.90e-02(+)	1.28e-01±6.60e-02(+)	9.57e-02±3.83e-02(+)	3.73e-02±1.83e-02
+/-	27/0/0	26/1/0	22/3/2	23/2/2	26/1/0	20/4/3	-
Friedman's rank	5.2593	6.0926	3.2778	4.1481	4.1296	3.537	1.5556

size and takes full advantage of the single-objective algorithm to search promising regions in high-dimensional space. In addition, the multiobjective algorithm balances the diversity so that MFMOEA finds better Pareto fronts than other compared algorithms. To observe the convergence of the comparison algorithms over a larger number of iterations, we computed the HV metrics for all the comparison algorithms after 500 iterations as shown in Table S-I of the Supplementary Material. According to results of Friedman's test, our algorithms still achieved the best performance. The convergence curves in Fig. S-I of the supplementary material demonstrate the ability of MFMOEA to quickly find promising regions.

C. Major Component Contribution Analysis

In this section, three components of MFMOEA are analyzed for their contribution. Two main components are: 1) the knowledge transfer strategy and 2) the diversity enhancement mechanism. They are separated from the original MFMOEA to obtain two incomplete variants of the algorithm. In addition, the proposed auxiliary task generation strategy is compared with two other state-of-the-art strategies to verify its effectiveness. To ensure the adequacy of the experiments, all variants of the algorithm are independently run 30 times on all 27 datasets. The mean and variance of the experimentally obtained HV values were then calculated. The results

of the rank sum test and Friedman's test are shown in Tables V and VI and S-IV of the supplementary material. The average running time for all variants of the algorithm over 30 runs is shown in Table S-III of the Supplementary Material. To visualize the performance of all variants of the algorithm, the distribution of nondominated solutions on nine representative datasets is plotted as shown in Figs. S-III and S-IV of the Supplementary Material.

1) *Verification of Knowledge Transfer Strategy:* As shown in Table V, two comparative tests are simultaneously compared with MFMOEA. MF-DE means that the diversity enhancement mechanism is retained while the knowledge transfer strategy is removed. This can be used to test whether the knowledge transfer strategy works. In addition, explicit, and implicit knowledge transfers are validated by adding them to a baseline algorithm (MF-DE), respectively. As shown in Table S-II of the Supplementary Material, the two knowledge transfer strategies can significantly improve the performance of the baseline algorithm on most datasets.

It is worth noting that an additional set of experiments using randomly generated complementary solutions should be set up as a control group. The above incomplete algorithm without knowledge transfer strategy is named MF-DR. As demonstrated by Table V, on 20 datasets, the two variant algorithms with the knowledge transfer strategy removed do not perform as well as MFMOEA. It is also seen that whether the diversity

TABLE IV
MEAN HV PERFORMANCE ON TEST DATA

Dataset	NSGA-II	MOEA/D	MOEA-D-PFE	DAEA	SM-MOEA	MFFS	MFMOEA
Abalone	5.38e-01±2.43e-03(+)	5.33e-01±2.34e-03(+)	5.40e-01±9.86e-04(+)	5.35e-01±0.00e+00(+)	5.29e-01±1.40e-03(+)	5.25e-01±2.26e-16(+)	5.42e-01±3.72e-04
Wine	8.24e-01±5.40e-03(+)	8.09e-01±1.34e-02(+)	8.43e-01±7.84e-03(+)	8.52e-01±6.33e-03(+)	8.44e-01±3.55e-03(+)	8.45e-01±5.63e-05(+)	8.94e-01±1.07e-02
Australian	6.22e-01±1.13e-16(+)	5.95e-01±3.60e-02(+)	6.30e-01±2.70e-03(+)	6.18e-01±8.20e-03(+)	6.13e-01±1.32e-02(+)	6.24e-01±3.55e-03(+)	6.46e-01±1.24e-02
Zoo	7.78e-01±1.94e-02(+)	7.53e-01±1.47e-02(+)	7.93e-01±6.26e-03(+)	7.69e-01±8.36e-03(+)	7.37e-01±1.65e-02(+)	7.97e-01±9.82e-03(+)	8.52e-01±1.75e-03
Climate	8.99e-01±7.92e-03(+)	8.69e-01±1.53e-02(+)	8.98e-01±4.82e-03(+)	8.86e-01±4.52e-03(+)	8.95e-01±4.61e-03(+)	8.72e-01±9.31e-05(+)	9.15e-01±2.74e-03
Vehicle	6.03e-01±5.72e-03(+)	6.10e-01±6.36e-03(+)	6.21e-01±4.46e-03(+)	5.89e-01±9.40e-03(+)	5.78e-01±1.42e-02(+)	6.07e-01±5.91e-03(+)	6.44e-01±6.85e-03
Hepatitis	7.39e-01±2.17e-02(+)	7.40e-01±2.87e-02(+)	7.97e-01±1.34e-02(+)	7.86e-01±1.53e-02(+)	7.82e-01±9.53e-03(+)	8.29e-01±2.26e-16(+)	8.45e-01±4.59e-03
SPECTF Heart	7.02e-01±9.06e-03(+)	7.05e-01±9.47e-03(+)	7.19e-01±6.98e-03(+)	7.43e-01±9.45e-03(-)	6.74e-01±1.29e-02(+)	7.10e-01±8.96e-03(+)	7.39e-01±5.05e-03
German	7.05e-01±4.43e-03(+)	6.96e-01±8.54e-03(+)	6.88e-01±1.13e-02(+)	7.12e-01±3.69e-03(+)	6.98e-01±5.16e-03(+)	7.01e-01±1.56e-03(+)	7.20e-01±6.91e-03
Flags	6.15e-01±1.36e-02(+)	5.65e-01±0.00e+00(+)	6.20e-01±1.57e-02(+)	5.69e-01±1.83e-02(+)	5.89e-01±1.65e-02(+)	6.48e-01±3.62e-03(+)	7.02e-01±2.49e-02
Ionosphere	9.13e-01±9.66e-03(+)	9.11e-01±1.24e-02(+)	9.28e-01±6.60e-03(+)	9.02e-01±1.10e-02(+)	9.08e-01±7.24e-03(+)	9.19e-01±1.05e-04(+)	9.30e-01±4.73e-03
Spambase	8.48e-01±1.08e-02(-)	7.95e-01±2.26e-16(+)	8.49e-01±5.76e-03(-)	8.09e-01±4.28e-03(+)	8.31e-01±1.05e-02(+)	8.58e-01±1.12e-03(-)	8.36e-01±3.54e-03
Sonar	8.15e-01±2.09e-02(+)	8.41e-01±1.02e-02(+)	8.57e-01±1.30e-02(+)	8.54e-01±2.63e-02(+)	8.38e-01±1.91e-02(+)	8.45e-01±1.86e-02(+)	8.63e-01±1.85e-02
Optical	8.79e-01±4.31e-03(=)	7.76e-01±2.26e-16(+)	8.57e-01±2.31e-03(+)	8.39e-01±2.68e-03(+)	8.63e-01±5.27e-03(+)	8.86e-01±4.18e-03(-)	8.82e-01±3.83e-03
Hill Valley	5.48e-01±1.13e-02(+)	5.10e-01±2.25e-02(+)	5.44e-01±8.76e-03(+)	5.33e-01±1.29e-02(+)	5.54e-01±8.31e-03(+)	5.91e-01±1.67e-02(+)	5.94e-01±9.07e-03
Arrhythmia	6.06e-01±1.08e-02(+)	5.70e-01±1.87e-02(+)	6.48e-01±1.46e-02(+)	6.29e-01±7.87e-03(+)	6.35e-01±1.83e-02(+)	6.22e-01±7.35e-03(+)	6.98e-01±1.08e-02
Madelon	6.07e-01±1.44e-02(+)	5.82e-01±2.32e-02(+)	6.68e-01±1.33e-02(+)	5.93e-01±8.29e-03(+)	7.51e-01±6.31e-02(+)	8.99e-01±2.89e-03(-)	8.80e-01±7.31e-03
Isotlet5	6.90e-01±1.03e-02(+)	6.62e-01±1.03e-02(+)	7.57e-01±5.66e-03(+)	7.32e-01±4.86e-03(+)	7.54e-01±1.21e-02(+)	8.18e-01±1.47e-02(-)	8.00e-01±1.84e-02
SRBCT	8.32e-01±6.53e-03(+)	8.12e-01±2.01e-02(+)	9.86e-01±8.81e-04(+)	9.72e-01±5.71e-03(+)	8.50e-01±6.03e-02(+)	9.33e-01±3.69e-02(+)	9.91e-01±2.66e-02
GLIOMA	6.88e-01±1.45e-03(+)	6.83e-01±1.47e-02(+)	9.34e-01±2.51e-02(-)	8.84e-01±1.63e-02(-)	8.66e-01±6.92e-02(+)	9.36e-01±5.00e-02(-)	8.72e-01±6.55e-02
Leukemia1	7.32e-01±3.34e-02(+)	7.31e-01±3.61e-02(+)	9.21e-01±2.67e-02(+)	8.66e-01±1.60e-02(+)	8.75e-01±5.90e-02(+)	9.27e-01±1.06e-02(+)	9.89e-01±3.07e-02
DLBCL	7.91e-01±1.36e-02(+)	7.62e-01±1.80e-02(+)	9.92e-01±1.12e-03(+)	9.81e-01±3.43e-03(+)	9.53e-01±3.86e-02(+)	9.91e-01±2.25e-02(+)	9.96e-01±2.28e-02
Brain1	6.04e-01±1.24e-02(+)	6.11e-01±4.52e-16(+)	7.64e-01±1.91e-02(+)	7.49e-01±5.84e-03(+)	7.52e-01±4.29e-02(+)	8.27e-01±3.80e-02(+)	8.41e-01±2.70e-02
Carcinom	6.81e-01±1.12e-02(+)	6.84e-01±1.13e-16(+)	8.55e-01±1.60e-02(+)	8.00e-01±1.24e-02(+)	7.97e-01±4.78e-02(+)	8.19e-01±3.67e-02(+)	9.08e-01±1.68e-02
Prostate	7.55e-01±2.48e-02(+)	8.50e-01±2.26e-16(+)	9.34e-01±1.83e-02(+)	9.11e-01±5.03e-03(+)	9.02e-01±3.64e-02(+)	9.29e-01±5.67e-02(+)	9.57e-01±4.21e-02
Leukemia2	6.78e-01±6.36e-02(+)	7.39e-01±3.39e-16(+)	9.34e-01±4.52e-02(+)	8.51e-01±5.48e-02(+)	8.58e-01±7.13e-02(+)	9.18e-01±9.55e-02(+)	9.54e-01±4.50e-02
CLLSUB	5.39e-01±2.06e-02(+)	5.77e-01±1.13e-16(+)	8.04e-01±3.49e-02(+)	7.10e-01±1.55e-02(+)	6.92e-01±1.04e-01(+)	7.55e-01±6.70e-02(+)	8.33e-01±4.36e-02
+/-	25/1/1	27/0/0	25/0/2	25/0/2	27/0/0	22/0/5	-
Friedman's rank	5.2222	6.2593	2.8889	4.4444	4.8889	2.9259	1.3704

enhancement mechanism is used to generate supplementary solutions does not significantly affect the effect of knowledge transfer. This shows that the knowledge transfer strategy is effective. In MFMOEA, the single-objective algorithm can usually find some valid feature combinations or information in the decision space. This knowledge can be transferred to the original multiobjective task to help it converge. In addition, the elite solutions in the original multiobjective task are also migrated to auxiliary single-objective tasks. It aims to guarantee the variety and advancement of solutions in auxiliary single-objective tasks. Therefore, the original multiobjective task can find a better Pareto front and have a higher HV value with the assistance of single-objective tasks. Table S-III and Fig. S-III in the supplementary material demonstrates that although MF-DR and MF-DE have shorter run times, they are not as good as MFMOEA at finding feature subsets with better classification performance.

2) *Verification of Diversity Enhancement Mechanism*: It can be found that the randomly generated supplementary solution operation is better than the diversity enhancement mechanism in terms of HV on seven datasets. This is because randomly generating feature subsets may explore some new feature interaction relations. Nevertheless, it can be seen from Table V that the effect of randomly generating supplementary solutions is not much more than that of our method. When the dimension of the dataset becomes larger, especially when

the number of dimensions is above five thousand, the diversity enhancement mechanism can better guide the population to converge and filter out redundant features. Therefore, the diversity enhancement mechanism is more appropriate. From Table VI, it can be clearly seen that the MF-KT algorithm, which removed the diversity enhancement mechanism, is significantly worse than MFMOEA on 24 datasets. This directly shows that the diversity enhancement mechanism is very effective. It is well known that it does not make sense to evaluate duplicate solutions. However, simply replacing them with randomly generated solutions does not work well. The reason is that historical search experience and knowledge are not considered to improve the search direction. Therefore, we solve this problem by exploiting new solutions obtained by local searches in promising regions of the decision space. In this way, the algorithm's ability to explore promising and unknown areas is fully developed. Fig. S-III in the supplementary material demonstrates that feature subsets found by MF-KT are worse than MFMOEA in terms of classification accuracy and size.

3) *Verification of the Proposed Auxiliary Task Generation Strategy*: In addition, the effectiveness of the proposed auxiliary task generation strategy is verified. By comparing with two other auxiliary task generation methods, the experimental results show that our designed auxiliary task generation strategy achieved best results and improved the success rate of

TABLE V
MEAN HV ON 27 DATASETS IN VALIDATION EXPERIMENTS OF
KNOWLEDGE TRANSFER STRATEGY

Dataset	MF-DR	MF-DE	MFMOEAD
Abalone	5.40e-01±1.23e-03(+)	5.40e-01±1.63e-04(+)	5.42e-01±3.72e-04
Wine	8.59e-01±2.95e-03(+)	8.85e-01±7.81e-03(+)	8.94e-01±1.07e-02
Australian	6.30e-01±5.45e-03(+)	6.36e-01±2.17e-03(+)	6.44e-01±6.85e-03
Zoo	8.06e-01±1.01e-02(+)	8.46e-01±0.00e+00(+)	8.52e-01±1.75e-03
Climate	9.14e-01±2.95e-03(=)	9.14e-01±1.06e-03(=)	9.15e-01±2.74e-03
Vehicle	6.09e-01±3.80e-03(+)	6.37e-01±4.89e-03(+)	6.44e-01±6.85e-03
Hepatitis	8.58e-01±1.03e-02(-)	8.26e-01±1.26e-02(-)	8.45e-01±4.59e-03
SPECTF Heart	7.43e-01±5.61e-03(-)	7.40e-01±7.34e-03(=)	7.39e-01±5.05e-03
German	7.35e-01±3.69e-03(-)	7.14e-01±4.68e-03(+)	7.20e-01±6.91e-03
Flags	7.13e-01±6.12e-03(-)	7.11e-01±7.64e-03(-)	7.02e-01±2.49e-02
Ionosphere	9.26e-01±4.63e-03(+)	9.26e-01±7.31e-03(+)	9.30e-01±4.73e-03
Spambase	8.46e-01±6.17e-03(-)	8.58e-01±2.46e-03(-)	8.36e-01±3.54e-03
Sonar	8.41e-01±9.98e-03(+)	8.47e-01±1.88e-02(+)	8.63e-01±1.85e-02
Optical	8.80e-01±2.34e-03(=)	8.78e-01±2.30e-03(+)	8.82e-01±3.83e-03
Hill Valley	5.76e-01±8.16e-03(+)	5.58e-01±1.04e-02(+)	5.94e-01±9.07e-03
Arrhythmia	6.85e-01±1.63e-02(+)	6.90e-01±1.47e-02(+)	6.98e-01±1.08e-02
Madelon	8.93e-01±3.43e-03(-)	8.93e-01±5.11e-03(-)	8.80e-01±7.31e-03
Isolet5	7.88e-01±1.28e-02(+)	7.98e-01±9.71e-03(=)	8.00e-01±1.84e-02
SRBCT	9.74e-01±4.02e-03(+)	9.80e-01±3.90e-03(+)	9.91e-01±2.66e-02
GLIOMA	9.34e-01±4.30e-02(-)	9.17e-01±5.15e-02(-)	8.72e-01±6.55e-02
Leukemia1	9.33e-01±2.36e-02(+)	9.72e-01±2.12e-02(+)	9.89e-01±3.07e-02
DLBCL	9.95e-01±1.40e-02(=)	9.93e-01±8.28e-03(=)	9.96e-01±2.28e-02
Brain1	7.77e-01±2.36e-02(+)	8.22e-01±2.18e-02(+)	8.41e-01±2.70e-02
Carcinom	8.91e-01±3.47e-02(+)	9.01e-01±2.70e-02(+)	9.08e-01±1.68e-02
Prostate	9.35e-01±2.17e-02(+)	9.47e-01±2.70e-02(+)	9.57e-01±4.21e-02
Leukemia2	9.38e-01±3.47e-02(+)	9.42e-01±1.98e-02(+)	9.54e-01±4.50e-02
CLLSUB	7.84e-01±3.95e-02(+)	8.30e-01±4.13e-02(=)	8.33e-01±4.36e-02
+/-/	17/3/7	17/5/5	-
Friedman's rank	2.3704	2.1481	1.4815

TABLE VI
MEAN HV ON 27 DATASETS IN VALIDATION EXPERIMENTS OF
DIVERSITY ENHANCEMENT MECHANISM

Dataset	MF-KT	MFMOEAD
Abalone	5.40e-01±2.92e-03(+)	5.42e-01±3.72e-04
Wine	8.72e-01±3.12e-03(+)	8.94e-01±1.07e-02
Australian	6.23e-01±2.39e-03(+)	6.44e-01±6.85e-03
Zoo	8.07e-01±9.31e-03(+)	8.52e-01±1.75e-03
Climate	9.09e-01±4.61e-03(+)	9.15e-01±2.74e-03
Vehicle	6.21e-01±4.66e-03(+)	6.44e-01±6.85e-03
Hepatitis	7.68e-01±2.05e-02(+)	8.45e-01±4.59e-03
SPECTF Heart	7.35e-01±9.44e-03(+)	7.39e-01±5.05e-03
German	7.16e-01±2.61e-03(+)	7.20e-01±6.91e-03
Flags	6.78e-01±1.39e-02(+)	7.20e-01±6.91e-03
Ionosphere	9.25e-01±1.01e-02(+)	9.30e-01±4.73e-03
Spambase	8.55e-01±3.37e-03(-)	8.36e-01±3.54e-03
Sonar	8.27e-01±2.22e-02(+)	8.63e-01±1.85e-02
Optical	8.77e-01±3.85e-03(+)	8.82e-01±3.83e-03
Hill Valley	5.71e-01±1.14e-02(+)	5.94e-01±9.07e-03
Arrhythmia	6.89e-01±8.69e-03(+)	6.98e-01±1.08e-02
Madelon	8.89e-01±4.77e-03(-)	8.80e-01±7.31e-03
Isolet5	7.84e-01±1.41e-02(+)	8.00e-01±1.84e-02
SRBCT	9.80e-01±4.03e-03(+)	9.91e-01±2.66e-02
GLIOMA	9.05e-01±2.68e-02(-)	8.72e-01±6.55e-02
Leukemia1	9.79e-01±3.84e-02(+)	9.89e-01±3.07e-02
DLBCL	9.91e-01±2.29e-02(+)	9.96e-01±2.28e-02
Brain1	7.60e-01±2.98e-02(+)	8.41e-01±2.70e-02
Carcinom	8.34e-01±2.28e-02(+)	9.08e-01±1.68e-02
Prostate	9.16e-01±3.80e-02(+)	9.57e-01±4.21e-02
Leukemia2	9.39e-01±7.55e-02(+)	9.54e-01±4.50e-02
CLLSUB	7.86e-01±3.74e-02(+)	8.33e-01±4.36e-02
+/-/	24/0/3	-

knowledge transfer compared with other strategies, as shown in Table S-IV and Fig. S-IV of the Supplementary Material.

D. Computation Time

To examine how well the suggested method performs in terms of computation time, we designed a comparison experiment based on the computation time of all compared algorithms on all datasets. In the experiment, we called 30 parallel cores to perform the experiment for each algorithm. The results of Friedman's test are listed in Table VII along with the average calculation time used by each compared algorithm over 30 separate runs.

In general, computing the classification performance of all feature subsets is one of the most extended parts of the time spent in wrapper-based methods. The learner needs to be trained with the data of selected features of all the samples in the training set to predict the sample's label. Therefore, the computation time of each algorithm is very long on three large sample size datasets Abalone, Spambase, and Madelon. In addition, as can be seen, MFMOEA has the shortest computation time on 15 datasets compared with all algorithms. It is worth noting that the top two algorithms with the shortest run times are decomposition-based algorithms (e.g., MOEA/D and

MFMOEA). There are two reasons why they are faster than other well-designed algorithms. First, DAEA, SM-MOEA, and MFPS all require the calculation of dominance relationships between multiple objective values. In contrast, dominance relationships are not computed in the environmental selection mechanism of the decomposition-based multiobjective algorithm. Second, all three advanced algorithms have complex distance or matrix update calculations, which causes them not to be fast.

Although MOEA/D, MOEAD-PFE, DAEA, and SM-MOEA have shorter training time than MFMOEA on the other 11 datasets, MFMOEA has achieved better classification accuracy than them. This shows that the knowledge transfer strategy and the diversity enhancement mechanism can help the original multiobjective task to find better feature subsets by using auxiliary single-objective tasks. In addition, the computation time of the first three compared algorithms is 5–10 times longer than that of the proposed algorithm on most datasets. The latter three compared algorithms take 2–8 times longer than the proposed algorithm on these datasets. These are especially true for high-dimensional datasets. It is because MFMOEA's knowledge transfer strategy enables the sharing of small and effective feature subsets found in

TABLE VII
AVERAGE TIME CONSUMED BY SEVEN ALGORITHMS ON 27 DATASETS (UNIT: s)

Dataset	NSGA-II	MOEA/D	MOEAD-PFE	DAEA	SM-MOEA	MFFS	MFMOEA
Abalone	744.54	2501.54	2088.03	235.63	5479.84	20636.94	2701.26
Wine	245.59	31.83	72.54	76.51	52.50	72.02	27.08
Australian	312.64	235.77	200.11	75.01	141.52	97.24	71.79
Zoo	468.19	130.55	101.08	64.72	67.68	42.57	26.79
Climate	583.25	136.46	166.11	85.41	79.84	241.02	76.53
Vehicle	8385.46	359.26	203.11	48.43	60.21	96.87	77.70
Hepatitis	410.27	127.20	162.72	527.41	302.92	57.12	79.26
SPECTF Heart	495.65	36.42	37.16	43.79	71.52	111.27	33.88
German	6875.26	659.02	257.42	64.36	215.22	134.48	189.68
Flags	563.94	68.28	96.11	103.78	386.97	74.38	49.57
Ionosphere	214.61	34.62	89.72	54.83	116.40	65.17	34.26
Spambase	3923.04	2644.48	3155.31	364.96	2049.75	21801.59	2501.48
Sonar	477.91	42.53	54.56	55.06	67.75	59.50	69.68
Optical	653.22	123.57	831.82	681.63	571.57	3457.27	341.92
Hill Valley	1192.55	746.22	176.54	146.81	64.19	97.31	81.31
Arrhythmia	380.55	137.40	149.76	165.83	45.04	99.15	61.65
Madelon	8236.26	1626.42	1496.05	3529.13	3204.86	4373.36	3016.56
Isolet5	5874.56	1355.24	724.79	1439.42	1766.96	2111.47	701.87
SRBCT	647.53	83.75	92.45	84.14	75.63	96.95	95.47
GLIOMA	689.37	200.39	163.62	124.92	177.29	146.98	164.49
Leukemia1	921.75	117.28	215.10	210.38	230.26	300.18	85.32
DLBCL	123.14	121.98	177.94	246.38	303.54	316.28	42.22
Brain1	1176.26	162.80	786.11	220.43	244.12	397.26	108.53
Carcinom	1662.55	252.16	616.93	556.53	433.98	1196.16	331.76
Prostate	2283.26	284.94	822.17	439.89	684.46	1445.83	171.83
Leukemia2	1285.16	246.28	577.89	523.89	729.80	1036.55	155.21
CLLSUB	3153.26	291.19	1280.69	394.18	836.00	1668.72	165.83
Friedman's rank	6.5185	3.2741	4.2963	3.5074	3.8889	4.7407	2.0741

auxiliary single-objective tasks. This reduces the evaluation time. In summary, in most cases, our algorithm can often discover feature subsets with good classification accuracy and fewer features quickly.

VI. CONCLUSION

In this article, a new framework was proposed to solve the multiobjective feature selection problem by using multiple single-objective tasks to assist an original multiobjective task. A knowledge transfer strategy utilizing different task search experiences was designed to help the original multiobjective task converge. We also found the duplicate solution problem and the missing Pareto front problem in the feature selection method. A diversity enhancement mechanism was designed, which improves the algorithm in the promising area exploration ability.

The proposed MFMOEA has been tested on 27 benchmark classification datasets and compared with six state-of-the-art algorithms. They are all classic algorithms or new algorithms from top journals. It is worth mentioning that the feature dimensions of these datasets are diverse, ranging from 8 to 11 340. The experimental results show that on most datasets, MFMOEA outperforms other compared algorithms in both HV and IGD metrics. Analyzing the nondominated front, feature subsets found by our algorithm have the best classification accuracy and the most diverse number of features. The reason is that the knowledge transfer strategy enables the algorithm to quickly find the potential promising region in the process of evolution. The diversity enhancement mechanism efficiently

explores more promising regions and finds diverse feature subsets. In general, the proposed method can successfully obtain good feature subsets for classification problems.

Although the proposed method is promising, it has drawbacks in calculation time. The following several improvements can be studied in the future. First, an effective initialization strategy can be designed to help the algorithm reduce the search time. In addition, it is worth trying to combine domain expertise (such as the medical cancer field) to help select some key features faster.

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